AYMANE AIT DADS

Research Intern - Image & Video Processing | Deep Learning · Computer Vision · Numerical Optimization

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT and discovering that reasoning over-expansion - not incorrect reasoning - was the primary failure mode across 5 structured categories. Achieved 1st place in the EURECOM class leaderboard (0.791 AUC) for AI-generated image detection using a CLIP ViT-L/14 backbone with ensemble TTA, and submitted results to the NTIRE 2026 @ CVPR CodaBench international benchmark. Brings proven depth in deep learning, computer vision, and image processing - the three core technical pillars of Disney Research Studio - alongside rigorous ablation methodology and a track record of significant results in international competitions. Aims to contribute to research in image, video, or geometry processing at Disney Research Zurich, translating experimental findings into communicable research agendas for senior researchers and cross-functional teams.

CORE COMPETENCIES

Deep Learning | Computer Vision | Image Processing | Numerical Optimization | Machine Learning | Software Engineering | Research & Development | Problem-Solving

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated **Random Forest + K-Means machine learning models** on 100K+ fiber-optic network records, classifying network quality across thousands of nodes using upload speed, latency, and jitter features.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows built in Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, communicating research project results to non-technical decision-makers.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that over-expanded reasoning - not incorrect reasoning - was the primary failure mode, and constructed custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

AI-Generated Image Detection — *EURECOM ImSecu · NTIRE 2026 @ CVPR CodaBench · 1st in Class Leaderboard*

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark, detecting AI-generated images using a fine-tuned CLIP ViT-L/14 backbone (428M parameters, pretrained on 400M image-text pairs) with a custom MLP head and 10-view TTA ensemble.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC)**, demonstrating superior out-of-distribution generalization - trained on 250K images spanning 25+ generator types with FP16 mixed precision and dual learning-rate scheduling.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

Computer Vision & Image Processing: PyTorch | CLIP ViT | CNN | DenseNet | OpenCLIP | FP16/BF16 Mixed Precision | TTA Ensembling

Deep Learning & ML Research: LoRA SFT | Hugging Face Transformers | Scikit-learn | Ablation Studies | Unsloth | PEFT | TRL

Engineering & Tools: Python | Git | Linux | Kaggle GPU | Docker | Power BI | SQL